

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

Degree, Semester& Branch: VII Semester B.E. ECE B

Course Code & Title: EC8701 Antennas and Microwave Engineering

Name of the Faculty member: Mr.P.Venkatesh

Name of the Topic: Amplifier Power equations

Name of the Innovative Practice: Mind map

ICT tool used: Whiteboard

Description:

Mind Map Technique:

A mind map is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

A mind map is a diagram for representing tasks, words, concepts, or items linked to and arranged around a subject using a non-linear graphical layout that allows the student to build an intuitive framework around a central concept.

Learning Outcomes:

1. The students will be able to differentiate the operating power gain, Available power gain, Transducer gain.
2. The students can be acquired better knowledge about the amplifier power relations.
3. The students can be easily identified the corresponding amplifier power relation equations for solving the problems.

Use of appropriate method:

Justification for choosing the Mind Map Activity:

Mind map activity gives the graphical representation of Amplifier power relations. It differentiates the Power gain, Available power gain and Transducer Power gain clearly which will be helpful for understanding the concepts of amplifier power relations to the students.

Effective presentation:

I have planned the activity for 15 minutes but students don't know how to do this activity. So, I have explained how to draw the mind map for this particular topic and also the benefits of mind map during examination. I encouraged the students which it takes another 5 minutes for completing the activity.

Reflective Critique:**Challenges:**

The students confused about the different types of amplifier power gains (Operating Power gain, Available Power Gain and Transducer Power gain) which was derived in the class room. After that, I have chosen Mind Map as an Innovative practice method for better understanding about Amplifier power relations.

Benefits:

The Students eagerly participated in that activity and then they acquired the concept of amplifier power relations. Due to this activity, the students were able to recall the topic and remember the important terms easily.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	3	2	1					2	1		

CO5: The students will be able to design a microwave system for the given application

References:

1. https://www.youtube.com/watch?v=V9S6o4tcc_I
2. Reinhold Ludwig and Gene Bogdanov, "RF Circuit Design: Theory and Applications

